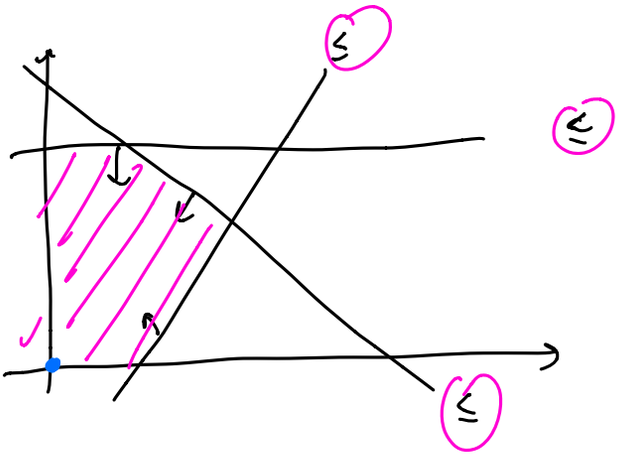
CHAPTER 3 – The Simplex Method

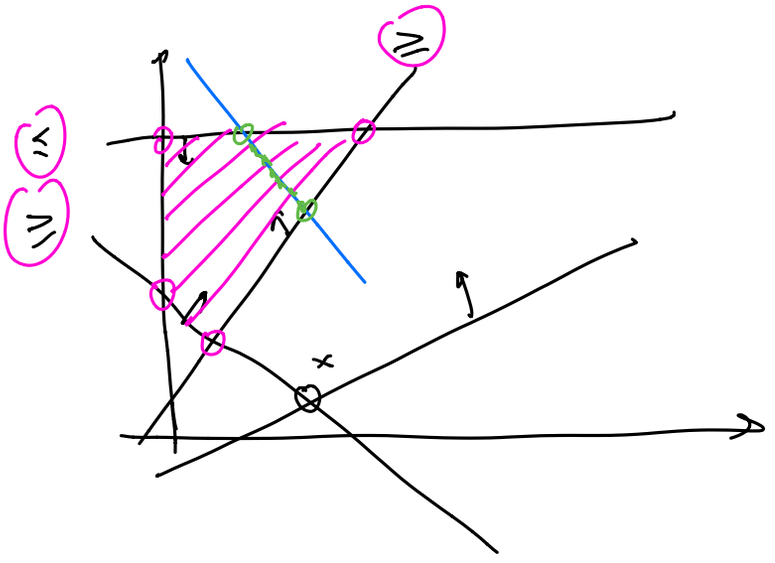
Artificial Starting Solutions

- It is important to have an identity matrix to start the simplex iterations.
- If all constraints are ≤ type, then the identity matrix is readily available using the slack variables.
- In case there are (=) and (≥) constraints starting is not that easy.
- At the initial phases artificial variables are used.
- There are two common methods: M-method (big M method)
  Two-phase method

LPs in which all the constraints are <= with nonnegative RHSs offer a convenient all-slack starting basic feasible solution (all non-slacks are set to zero if you remember the example i.e. $(x_1,x_2)=(0,0)$.
But models having >= and/or = do not.
In these models the origin point is not guaranteed to be a feasible point to start.

In this case, we use artificial variables that play the role of slacks in the first phase. Then, they are disposed and we continue to Phase II if Phase I ends successfully.
We add an artificial variable to all = and >= equations as we illustrate in the following example.





Two-Phase Method

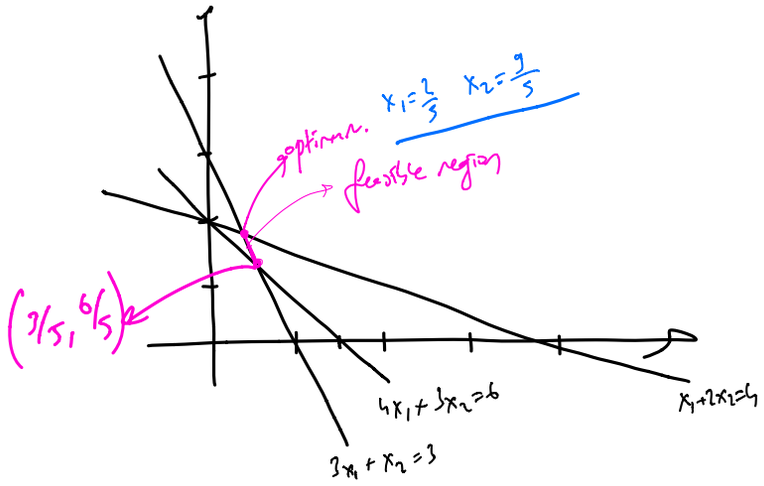
Phase 1: Put the problem in equation form
Add the necessary artificial variables
Find a basic solution which minimizes the sum of artificial variables
→ If the sum is positive stop, LP is infeasible
→o/w proceed to phase 2

Phase 2: Use the feasible solution from phase 1 as the starting basic feasible solution for the original problem.

Example:

$$\min z = 4x_1 + x_2$$
$$s.t. \quad 3x_1 + x_2 = 3 \quad !$$
$$4x_1 + 3x_2 \geq 6 \quad !$$
$$x_1 + 2x_2 \leq 4 \quad \checkmark$$
$$x_1, x_2 \geq 0$$

→



$$\min \quad 4x_1 + x_2$$
$$3x_1 + x_2 \geq 3$$
$$4x_1 + 3x_2 - x_3 = 6$$
$$x_1 + 2x_2 + x_4 = 4$$
$$x_1, x_2, x_3, x_4 \geq 0$$

⇓

minimize $r = R_1 + R_2$

$$3x_1 + x_2 + R_1 = 3$$
$$4x_1 + 3x_2 - x_3 + R_2 = 6$$
$$x_1 + 2x_2 + x_4 = 4$$
$$x_1, x_2, x_3, x_4, R_1, R_2 \geq 0$$

Basic	x_1 ↓	x_2	x_3	R_1	R_2	x_4	sol	MR
r	0 4	0	0	-1	-1	0	0	
← R_1	3	1	0	1	0	0	3	← 3/3
R_2	4	3	-1	0	1	0	6	6/4
x_4	1	2	0	0	0	1	4	4/1

we have to make the artificial variables zero (i.e get rid of them). Therefore, we apply the simplex operations to make them nonbasic variables.

Basic	x_1	x_2 ↙	x_3	R_1	R_2	x_4	sol	MR
r	0	0	0	-1	-1	0	0	
x_1	1	1/3	0	1/3	0	0	1	1/1/3 = 3
← R_2	0	5/3	-1	-4/3	1	0	2	← 2/5/3 = 1.2
x_4	0	5/3	0	-1/3	0	1	3	3/5/3 = 1.8
v	0	0	0	-1	-1	0	0	
x_1	1	0	1/5	3/5	-1/5	0	3/5	
x_2	0	1	-3/5	-4/5	3/5	0	6/5	
x_4	0	0	1	1	-1	1	1	

Observe that R_1, R_2 are nonbasic (i.e they are zero).

observe that r is 0.

r should be nonpositive to proceed to phase II.

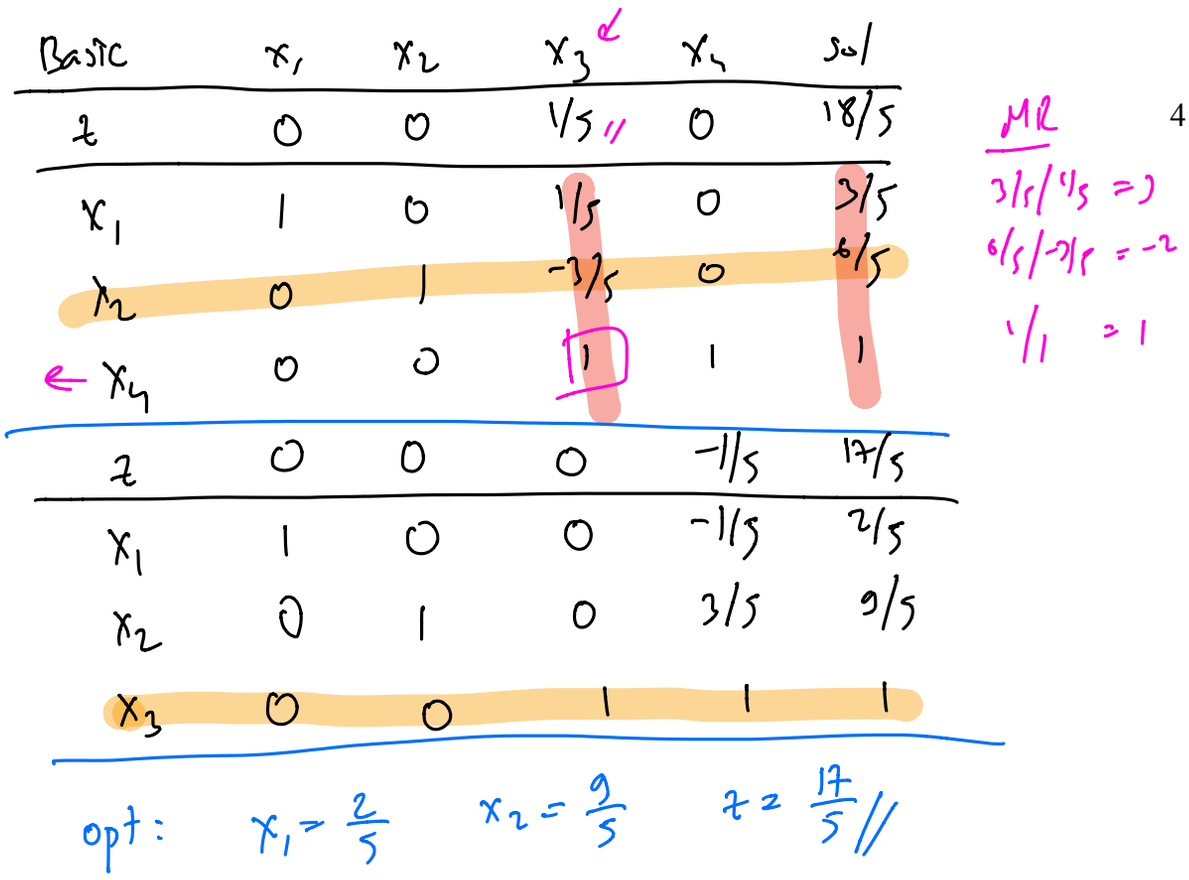
Minimize $z = 4x_1 + x_2$

$$x_1 + 1/5\, x_3 = 3/5$$
$$x_2 - 3/5\, x_3 = 6/5$$
$$x_3 + x_4 = 1$$
$$x_1, x_2, x_3, x_4 \geq 0$$

Basic	x_1	x_2	x_3	x_4	sol
z	-4	-1	0	0	0
4X → x_1	1	0	1/5	0	3/5
1X → x_2	0	1	-3/5	0	6/5
x_4	0	0	1	1	1

Because the basic variables x_1 and x_2 have nonzero coefficients in the z-row, they must be substituted out i.e their coeff. must be made zero.

new z-row = old z-row + (4 X x_1-row + 1 X x_2-row)



At the end of phase 1, artificial variable columns are deleted if they are all non-basic.

If an artificial variable is basic at zero level, it transfers to phase 2 as a part of the starting solution.

In such a case, that artificial variable should be forced to leave basis. It should be selected as the leaving variable even if its pivot column coefficient is negative or it is not the MR.

Special Cases in Simplex

Degeneracy
Alternative optima
Unbounded solution
Infeasibility

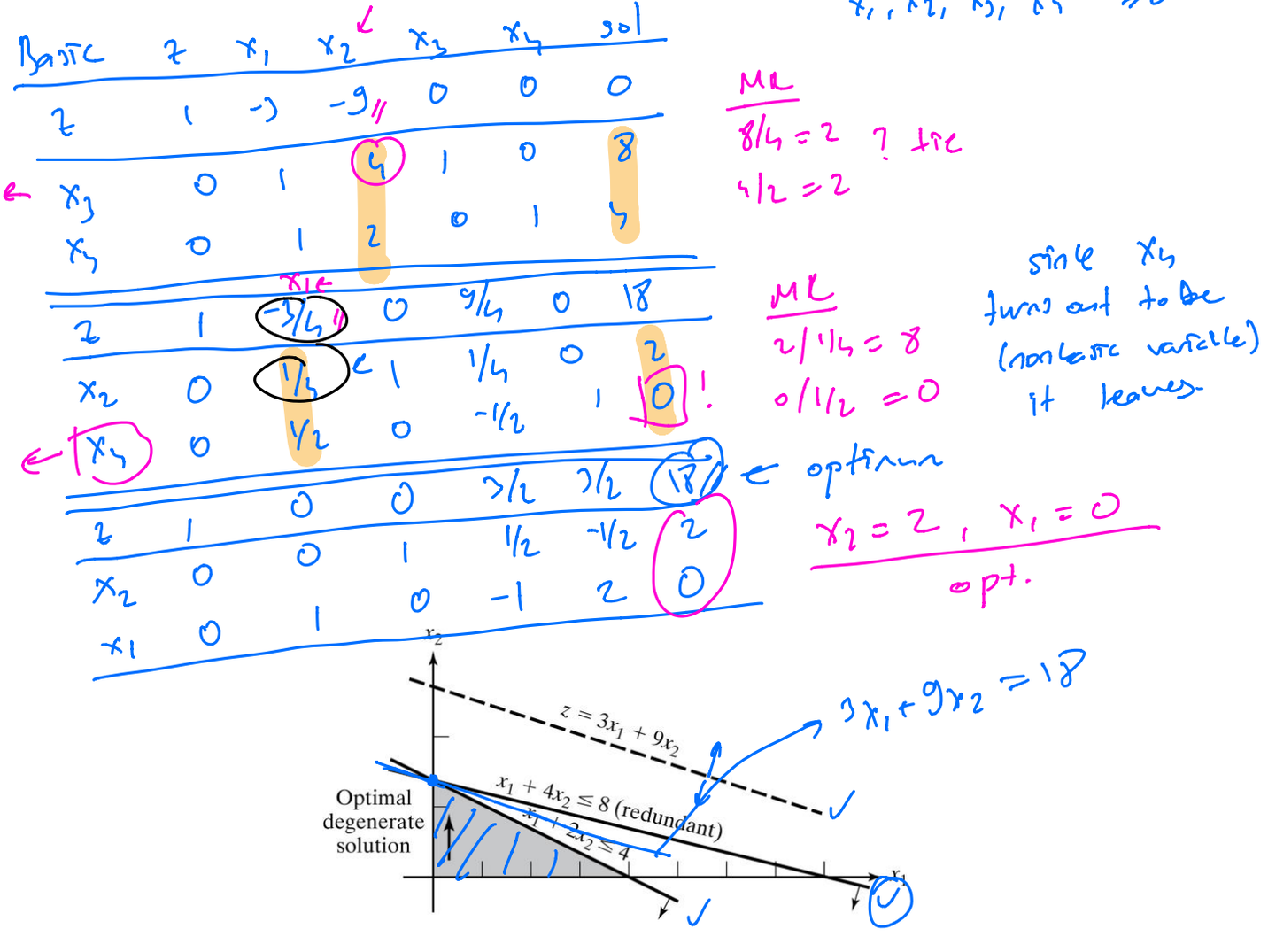
Degeneracy

If a tie occurs for MR, in the next iteration at least one basic variable will be zero. Such a solution is said to be degenerate

→ There is at least one redundant constraint
→ There could be cycling in simplex (However, its probability is small)

- There are methods to prevent cycling but very time consuming. Thus, usually it is not implemented.

$$\max \quad z = 3x_1 + 9x_2$$
$$x_1 + 4x_2 \le 8$$
$$x_1 + 2x_2 \le 4$$
$$x_1, x_2 \ge 0$$



max $3x_1 + 9x_2$
$x_1 + 4x_2 + x_3 = 8$
$x_1 + 2x_2 + x_4 = 4$
$x_1, x_2, x_3, x_4 \ge 0$

MR
$8/4 = 2$? tie
$4/2 = 2$

MR
$2/1/4 = 8$
$0/1/2 = 0$

since x_4 turns out to be (nonbasic variable) it leaves.

optimum
$x_2 = 2, \; x_1 = 0$
opt.

$3x_1 + 9x_2 = 18$

Alternative Optima

If the objective function lines are parallel to a binding constraint then, there are infinite numbers of alternative solutions with the same objective value.
Alternative optima can be detected in simplex if the z-row coefficient of a non-basic variable is zero. That means it can enter the basis without changing the objective value.

$$\begin{aligned} \max \quad & z = 2x_1 + 4x_2 \\ & x_1 + 2x_2 \le 5 \\ & x_1 + x_2 \le 4 \\ & x_1, x_2 \ge 0 \end{aligned}$$

$$\begin{aligned} \max \quad & 2x_1 + 4x_2 \\ & x_1 + 2x_2 + x_3 = 5 \\ & x_1 + x_2 + x_4 = 4 \\ & x_1, x_2, x_3, x_4 \ge 0 \end{aligned}$$

Basic	x_1	x_2	x_3	x_4	sol
z	-2	-4	0	0	0
x_3	1	2	1	0	5
x_4	1	1	0	1	4
z	0	0	2	0	10
x_2	1/2	1	1/2	0	5/2
x_4	1/2	0	-1/2	1	3/2
z	0	0	2	0	10
x_2	0	1	1	-1	1
x_1	1	0	-1	2	3

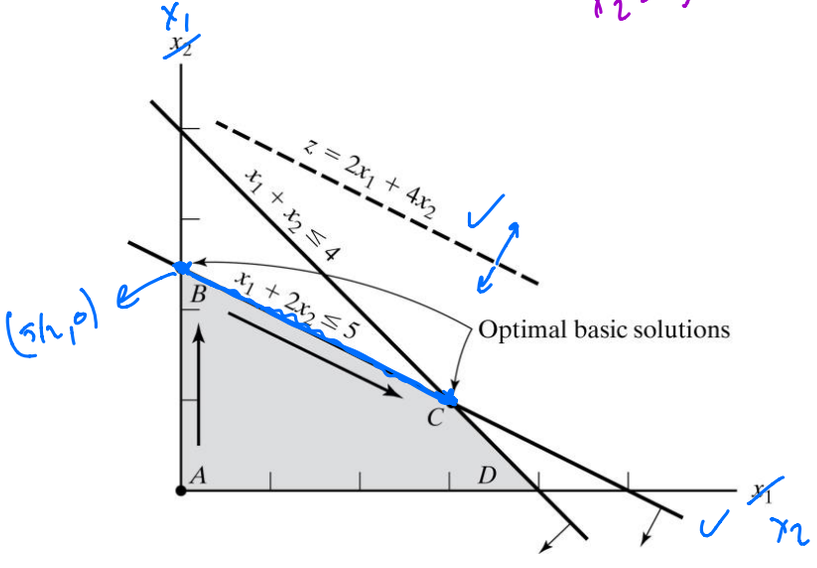
MR: $5/2 = 2.5$, $4/1 = 4$

MR: $5/2 \div 1/2 = 5$, $3/2 \div 1/2 = 3$

opt: $x_1 = 5/2$, $x_2 = 0$, $x_4 = 3/2$

The zero coefficient of nonbasic x_1 indicates that x_1 can be made basic altering the values of the basic variables without changing the value of z.

$x_1 = 3$, $x_2 = 1$ → (C)



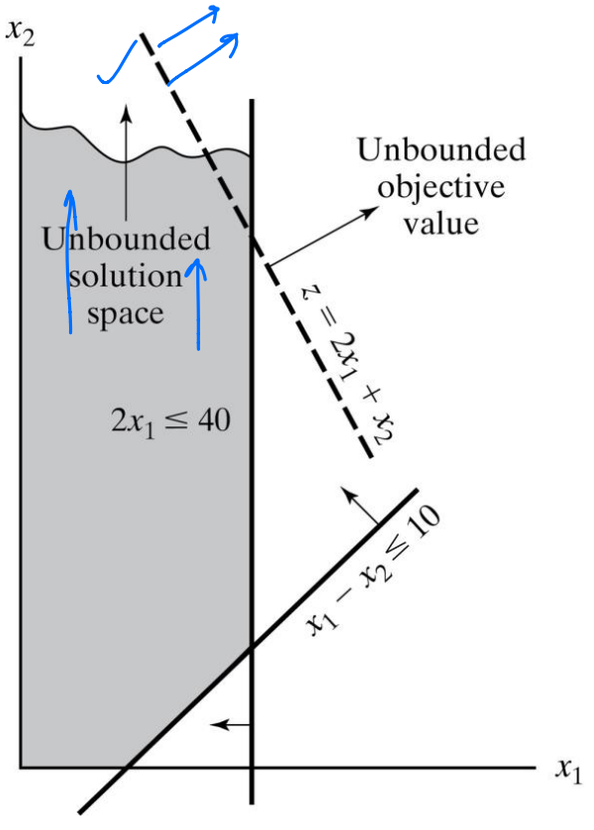
(5/2, 0)

Unboundedness

$$\begin{aligned} \max \quad & z = 2x_1 + x_2 \\ & x_1 - x_2 \le 10 \\ & 2x_1 \le 40 \\ & x_1, x_2 \ge 0 \end{aligned}$$

Basic	x1	x2	x3	x4	solution
z	-2	-1	0	0	0
x3	1	-1	1	0	10
x4	2	0	0	1	40

- If the constant coefficients of any non-basic variable are all non-positive then, the feasible region is unbounded.
- If in addition the z-row coefficient of that variable is negative (in max. problem) or positive (in min. problem) then, the objective value is unbounded too.



Infeasibility

If the constraints are inconsistent then, there is no feasible solution of the problem.
Infeasibility can be detected if any artificial variable assumes a positive value at the final iteration of simplex. (both M-method and two-phase method)

